

SPS Student Cover Sheet / Feedback Sheet

***Indicated fields MUST be completed by student.**

*Exam number	B141395
*Course	<u>Anthropology and Environment</u>
*Essay title	Environmental and Social Impact assessment for implementation of EGG lighting's 'Lighting as a Service model' in Scotland
*Word count	987
Marker's name	

*If you consent for the School to retain a copy of your coursework and make it available to other students to assist them in completing their assignment, please check the box below. Where your work is made available, it will be anonymized. Please be sure to read the full details of our coursework retention policy [here](#) before providing your consent.

I consent to my coursework being retained for the purpose stated above ☐.

NOTE: Essay marks are reviewed by another member of staff prior to being returned to students

This Section is for office use.

Initial Mark	
Word count penalties	
Lateness penalties	
Adjusted Mark	

Overview

Aspect of performance	+		Avg		-
<i>Thinking skills</i> (criticism, analysis, interpretation, logic, argumentation, evaluation, use of comparison, anticipating counter-arguments, etc.)					
<i>Comprehension</i> (accuracy in facts, details and representation of author's views, breadth of reading, grasp of major issues, etc.)					
<i>Writing skills</i> (structure and organisation, clarity, precision, grammar/spelling, referencing, use of illustration, style, etc.)					

Major advice to student

Main strength(s) of the essay	
Main weakness(es) of the essay	
This and future essays could be improved by...	

Environmental and Social Impact assessment for implementation of EGG lighting's 'Lighting as a Service model' in Scotland

Executive summary

EGG Lighting is a Glasgow-based lighting manufacturer that has been developing a 'Lighting as a Service' (LaaS) model with help from Zero Waste Scotland, the European Regional Development Funds and the Scottish Government. It is a circular economy business model that would allow customers to pay and receive a lighting service rather than owning the fittings and bulbs. It can provide EGG Lighting a steady, long-term income flow and ongoing relationship with customers, while the customer has access to the latest LED technology at an affordable cost.

LaaS would provide end users the benefits of the latest technology without them having to completely replace their lighting system frequently, minimizing the waste generated while reducing energy consumption. In traditional linear economy lighting industries, old lighting systems passed its expectancy or needed upgrade will be disposed of, when only 5% of the light fitting actually needs to be replaced. Instead of disposing the whole lighting system unit, EGG Lighting's new modular approach to lighting systems allows them to only replace the 5% necessary. By minimizing materials waste and extending the life of lighting products, this project hopes to reduce environmental impact caused by the current linear lighting industry.

Project – background information

The adoption of circular economy principles by EGG Lighting allows users to pay only for the light as a service, and not for the equipment. This provides the developers incentive to remove built-in obsolescence in their lighting products and design their products in a fully sustainable way as they retain ownership of the lighting system.

LaaS had already been proven successful in many countries outside Scotland, The National Flight Academy took advantage of Lighting-as-a-Service to retrofit their 102,000 square foot facility in Pensacola, Florida in 2017 and they were able to get a 54% wattage reduction immediately and an estimated \$204,804USD in net savings over 10 years(FSG Solutions, 2017).

Rapid improvements in luminaire technologies in recent years have greatly improved their efficiency, but while upgrading lighting systems to the latest models may make good environmental and economic sense, constant upgrades can be expensive to the end user and wasteful for the supplier. In the UK, 48.5% of lamps are recycled, with 6,132 tonnes of waste collected for recycling (*Zero Waste Scotland*, 2021). However, most of the time only 5% of the lighting unit actually needs to be replaced, meaning 95% of the remaining material is unnecessarily discarded (*Cole*, 2018). EGG Lighting has developed a new modular approach to lighting systems, which allow the LED and driver parts to be replaced separately while leaving the rest of the unit in place, greatly reducing the amount of waste generated. The 'lighting as a service' model would allow users to continue to receive regular upgrades to the lighting technology, but without having to dispose of the entire old unit and constantly paying for new lights.

Project impacts

Environmental

Positive:

- As providers retain ownership of the lighting systems, planned obsolescence becomes an undesirable design, lightings will also be designed in a fully sustainable way, greatly reducing the amount of waste it generates.
- The lighting products themselves will also get more and more efficient after every upgrade, further reducing energy consumption and thus emissions.

Negative:

- Increased logistics due to frequent upgrade and maintenance of still serviceable lighting systems might lead to extra carbon emissions.

Social and Economic

Positive:

- The subscription to LaaS allows customers to get exclusive use of the latest LED technology and to treat them as their own but without any of the ownership responsibilities and disposal or recycling concerns, with a much lower upfront cost than buying the lights themselves. This subsequently has a positive impact on the end users as it eliminates risks of unforeseeable damage or misuse.
- When new LED technology comes out, EGG Lighting will upgrade the lights to match it at no additional cost. The newest LED technology would also mean better efficiency and cheaper bills after every upgrade.
- Cash Flow is more easily managed under the 'LaaS' model because the monthly payments for the service are fixed well in advance and will only get cheaper after every upgrade.
- Constant energy savings leads to lower cost for lighting.

Negative:

- The subscription to 'LaaS' model does not involve the customer's acquisition of the lighting product itself and the provider retains the ownership of the product.

- Subscription based services are often hard to cancel, customers will be bound to its provider and held to the terms of the agreement for the duration of the contract.
- Frequent updates of lighting systems and scheduled maintenance can cause inconvenience for the customers. And for EGG Lighting, these maintenance and upgrades can be very costly and difficult to plan out especially when there is a large number of household customers scattered around the country.
- As EGG Lighting is the first and only lighting company providing LaaS in Scotland at the moment, customers interested in LaaS don't have a choice besides EGG lighting as of now, but more lighting providers are expected to follow suit in the future.

Recommendations

The global lighting as a service market and luminaire technologies are growing rapidly as it has no apparent drawbacks especially for businesses as the opportunity to capture significant energy savings sooner is a huge financial benefit. However, popularising such a model amongst the general public would be hard as it is very labour intensive to fit and maintain lights in every single household. EGG Lighting should work with the Scottish Government educating the public benefits of LaaS, making it the mainstream lighting products and replacing conventional lighting products as it is a collective effort to reduce the environmental impact caused by lighting products. The Scottish Government should also start making regulations for such a novelty as EGG Lighting is the sole company providing LaaS in Scotland at the moment.

References cited

EGG Lighting. n.d. *EGG Circular* | *EGG Lighting*. [online] Available at: <<https://www.egglighting.com/egg-circular>> [Accessed 5 February 2021].

Cole, R., 2018. "*Scottish lighting company develops circular LED business model.*" [online] Resource Magazine. Available at: <<https://resource.co/article/scottish-lighting-company-develops-circular-led-business-model-12985>> [Accessed 7 February 2021].

Zero Waste Scotland. 2021. "*EGG Lighting.*" [online] Available at: <<https://www.zerowastescotland.org.uk/case-study/egg-lighting#:~:text=As%20part%20of%20this%2C%20the,owning%20the%20fitting%20and%20bulbs.&text=The%20model%20calculates%20contract%20rates,service%20to%20EGG's%20target%20customers.>> [Accessed 5 February 2021].

ID: 4600423 2018. "*Global Lighting as a Service Market: Focus on Applications, Leading Players Ranking, and Competitive Landscape - Analysis and Forecast (2018-2025)*". researchandmarkets.com. Research And Markets. [Accessed 8 February 2021].

FSG Solutions. 2017. *LaaS Accelerates National Flight Academy's Investment in LED Lighting Technology.* [online] Available at: <https://nalmco.org/NALMCO/LM_M/Articles/LaaS_Accelerates_National_Flight_Academy_Investment_in_LED_Lighting_Technology.aspx> [Accessed 10 February 2021].